


Problem 1

A simple example: $\alpha = 1$, $Z = \{a, b, c\}$ $a < b < c$

$$v(a) = 2$$

$$v(b) = 3$$

$$v(c) = 4$$

$$\text{Lotteries } p = \left(\begin{matrix} \Pr(a) \\ 1/2 \end{matrix}, \begin{matrix} \Pr(b) \\ 0 \end{matrix}, \begin{matrix} \Pr(c) \\ 1/2 \end{matrix} \right) \rightarrow U(p) = \max \left\{ \frac{1/2}{2}, \frac{0}{3}, \frac{1/2}{4} \right\} = \frac{1}{4}$$

$$q = \left(\begin{matrix} 1/8 \\ 3/4 \end{matrix}, \begin{matrix} 1/8 \end{matrix} \right) \rightarrow U(q) = \max \left\{ \frac{1/8}{2}, \frac{3/4}{3}, \frac{1/8}{4} \right\} \\ = \max \left\{ \frac{1}{16}, \frac{1}{4}, \frac{1}{32} \right\} = \frac{1}{4}$$

$$U(p) = U(q) \Leftrightarrow p \sim q$$

$$w = \frac{1}{2}p \oplus \frac{1}{2}q = \left(\frac{5}{16}, \frac{3}{8}, \frac{5}{16} \right) \rightarrow U(w) = \max \left\{ \frac{5/16}{2}, \frac{3/8}{3}, \frac{5/16}{4} \right\}$$

$$= \max \left\{ \frac{5}{32}, \frac{1}{8}, \frac{5}{64} \right\} = \frac{5}{32}$$

$$(a) p \sim q, \quad p = \frac{1}{2}p \oplus \frac{1}{2}p \text{ vs } \frac{1}{2}q \oplus \frac{1}{2}p = w$$

$$U(p) = \frac{1}{4} > U(\frac{1}{2}q \oplus \frac{1}{2}p) = U(w) = \frac{5}{32}$$

$\therefore$ Independence fails.

(a) Continuity: $U(p) = \alpha \max \left\{ \frac{p(z)}{v(z)} \mid z \in Z \right\}$

$\frac{p(z)}{v(z)}$ is continuous for all $z \in Z$
because $v(z)$ is continuous and $v(z) > 0 \forall z \in Z$

Since $\max \left\{ \frac{p(z)}{v(z)} \mid z \in Z \right\}$ is continuous in $\frac{p(z)}{v(z)}$, U is continuous.

(b) For any lottery p , How $U([E_p])$ vs $U(p)$

$a < c$, Let $p = \lambda [c] \oplus (1-\lambda) [a] \rightarrow E_p = \lambda c + (1-\lambda)a$

$U(p) = \alpha \max \left\{ \frac{\lambda}{v(c)}, \frac{1-\lambda}{v(a)} \right\}$ $[E_p] = [\lambda c + (1-\lambda)a]$

$U([E_p]) = \alpha \max \left\{ \frac{1}{v(\lambda c + (1-\lambda)a)} \right\} = \frac{\alpha}{v(\lambda c + (1-\lambda)a)}$

Depending λ , $a, c, v(a), v(c)$
and $v(\lambda c + (1-\lambda)a)$
 $a < \lambda c + (1-\lambda)a < c$
 $v(a) < v(\lambda c + (1-\lambda)a) < v(c)$

Problem 2

p - price of y

w_1 - prices of inputs, w_1 is random
 w_2

$$TC = y^2 (w_1 + w_2)$$

$$\max_{y \geq 0} \mathbb{E} [p \cdot y - y^2 (w_1 + w_2)] = \max_{y \geq 0} \left\{ p \cdot y - y^2 (\mathbb{E}(w_1) + w_2) \right\}$$

$$\text{F.O.C.} \quad p - 2y^* (\mathbb{E}(w_1) + w_2) = 0 \Rightarrow y^* = \frac{p}{2(\mathbb{E}(w_1) + w_2)}$$

$$\text{S.O.C.} \quad -2(\mathbb{E}(w_1) + w_2) < 0 \quad \text{as } \text{leneg} \quad \nearrow \text{true max}$$

$\hookrightarrow \mathbb{E}(w_1) > 0$
 $w_1 > 0$

$$(a) \quad \mathbb{E}(w_1) = \theta w_{1,n} + (1-\theta) w_{1,e}$$

$$(b) \quad \mathbb{E}(w_1) = \int_{w_{1,e}}^{w_{1,n}} w_1 g(w_1) dw_1$$

Problem 3

(a) $X_1, \dots, X_K$ are returns and $\alpha = (\alpha_1, \dots, \alpha_K)$ are shares
 $0 \leq \alpha_i \leq 1$

$\sum_{i=1}^K \alpha_i X_i$ = Value of portfolio / dollar invested

$(\sum_{i=1}^K \alpha_i X_i) W$ = Total return (wealth)

Investors seeks to $\max_{\alpha = (\alpha_1, \dots, \alpha_K)} \int u((\sum_{i=1}^K \alpha_i X_i) W) dF(x_1) dF(x_2) \dots dF(x_K)$

s.t. $0 \leq \alpha_i \leq 1$ $\left\{ \begin{array}{l} 0 \leq \alpha_i \leq 1 \quad i=1, \dots, K \\ \sum_{i=1}^K \alpha_i \leq 1 \end{array} \right.$

$\hookrightarrow \int u((\sum_{i=1}^K \alpha_i X_i) W) f(x_1) f(x_2) \dots f(x_K) dx_1 \dots dx_K$

(b) $K=2$
 For any $0 \leq \alpha \leq 1$, define the
 value of the portfolio given α as

$$Y_\alpha = \alpha X_1 + (1-\alpha) X_2, \quad Y_{1/2} = \frac{1}{2} X_1 + \frac{1}{2} X_2$$

I will show that Y_α is a mean-preserving spread of $Y_{1/2}$

$$Y_\alpha = \alpha X_1 + (1-\alpha) X_2 + \underbrace{\frac{1}{2} X_1 + \frac{1}{2} X_2}_{Y_{1/2}} - \frac{1}{2} X_1 - \frac{1}{2} X_2$$

$$Y_\alpha = \alpha X_1 + (1-\alpha) X_2 + Y_{1/2} - \frac{1}{2} X_1 - \frac{1}{2} X_2$$

$$Y_\alpha = Y_{1/2} + \underbrace{(\alpha - \frac{1}{2}) X_1 + (1 - \alpha - \frac{1}{2}) X_2}_{\mathcal{E}}$$

$$\begin{aligned} \mathcal{E} &= (\alpha - \frac{1}{2}) X_1 + (1 - \alpha - \frac{1}{2}) X_2 \\ &= (\alpha - \frac{1}{2}) X_1 - (\alpha - \frac{1}{2}) X_2 \\ &= (\alpha - \frac{1}{2}) (X_1 - X_2) \end{aligned}$$

$$E[\mathcal{E} | Y_{1/2}] = 0 \Leftrightarrow E[(\alpha - \frac{1}{2}) X_1 + (1 - \alpha - \frac{1}{2}) X_2 | Y_{1/2}] =$$

$$E[(\alpha - \frac{1}{2}) X_1 + (1 - \alpha - \frac{1}{2}) X_2 | \frac{1}{2} X_1 + \frac{1}{2} X_2] =$$

$$E[(\alpha - 1/2)(X_1 - X_2) \mid \frac{1}{2}(X_1 + X_2)] = (\alpha - 1/2) E[X_1 - X_2 \mid \frac{1}{2}(X_1 + X_2)] = 0$$

(c) $\alpha = (\alpha_1, \dots, \alpha_k)$

$$Y_\alpha = \sum_{i=1}^k \alpha_i X_i, \quad Y_{1/k} = \frac{1}{k} \sum_{i=1}^k X_i$$

$$Y_\alpha = Y_{1/k} + \sum_{i=1}^k (\alpha_i - 1/k) X_i$$

If we manage to show that $E\left(\sum_{i=1}^k (\alpha_i - 1/k) X_i \mid Y_{1/k}\right) = 0$

we are done.

$$E\left(\sum_{i=1}^k (\alpha_i - 1/k) X_i \mid Y_{1/k}\right) = \sum_{i=1}^k (\alpha_i - 1/k) E[X_i \mid Y_{1/k}] \stackrel{*}{=} \bar{A} \sum_{i=1}^k (\alpha_i - 1/k) = \bar{A} (1-1) = 0$$

* Since X_i is independent of X_j and identically distributed $\Rightarrow E(X_i \mid Y_{1/k}) = E(X_j \mid Y_{1/k}) = \bar{A}$
 ($\bar{A}$ is the common value $E[X_i \mid Y_{1/k}]$)